

Kabul University
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JavaScript Lecture#3rd

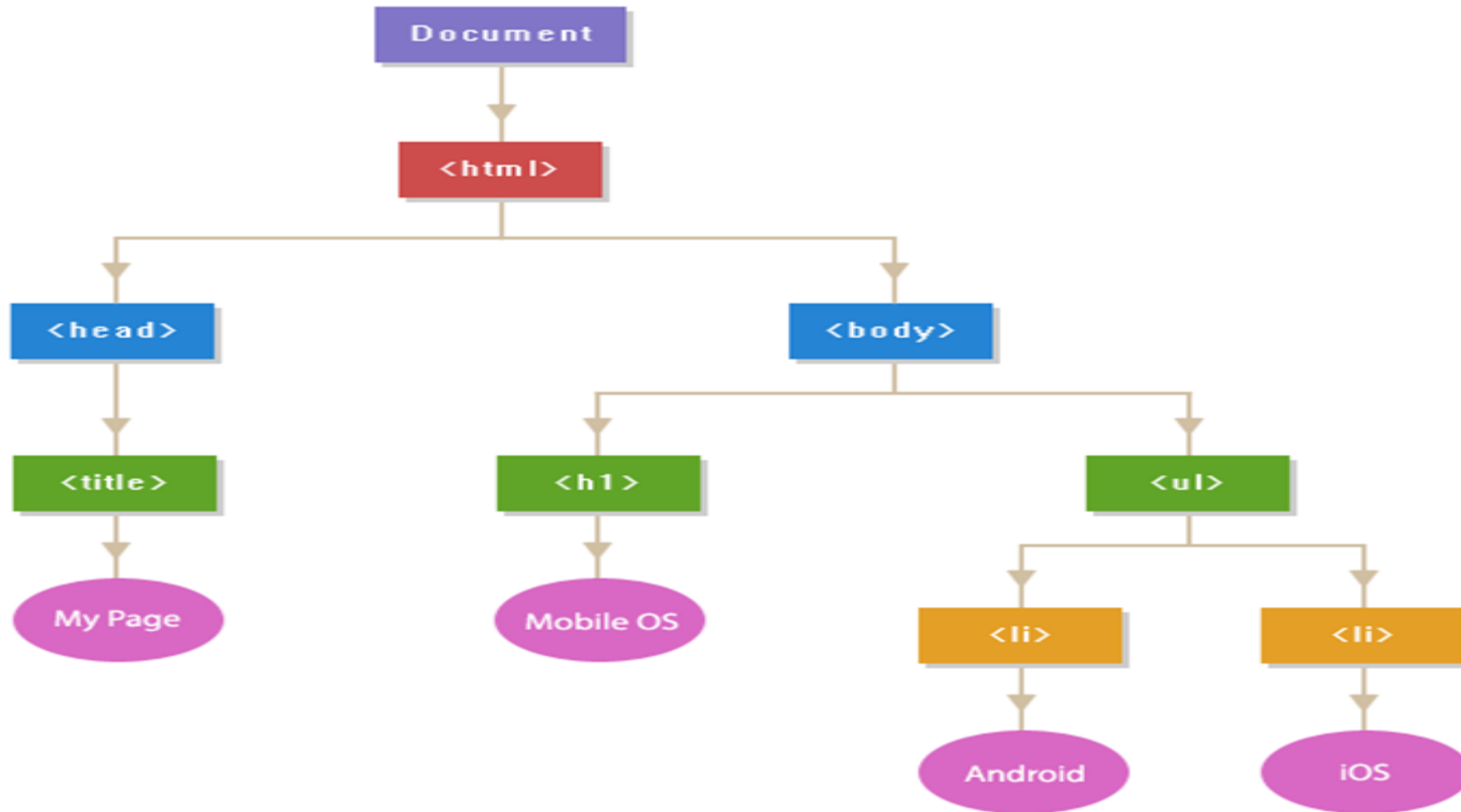
JavaScript DOM

- The Document Object Model, or DOM for short, is a platform and language independent model to represent the HTML or XML documents.
- It defines the logical structure of the documents and the way in which they can be accessed and manipulated by an application program.
- In the DOM, all parts of the document, such as elements, attributes, text, etc. are organized in a hierarchical tree-like structure; similar to a family tree in real life that consists of parents and children *nodes*.

JavaScript DOM

```
<!DOCTYPE html>
<html>
<head>
  <title>My Page</title>
</head>
<body>
  <h1>Mobile OS</h1>
  <ul>
    <li>Android</li>
    <li>iOS</li>
  </ul>
</body>
</html>
```

JavaScript DOM



- <html>
- <body>
- <p> this is paragraph</p>
- <h1> this is h1</h1>
-
- one
-
- </body>
- </html>

Methods of document object

• Method	Description
• <code>write("string")</code>	writes the given string on the document.
• <code>getElementById()</code>	returns the element having the given id value.
• <code>getElementsByName()</code>	returns all the elements having the given name value.
• <code>getElementsByTagName()</code>	returns all the elements having the given tag name.
• <code>getElementsByClassName()</code>	returns all the elements having the given class name .

Properties of document object

- **innerHTML**
- The **innerHTML** property can be used to write the dynamic html on the html document. It is used mostly in the web pages to generate the dynamic html such as registration form, comment form, links etc.
- **innerText**
- The **innerText** property can be used to write the dynamic text on the html document. Here, text will not be interpreted as html text but a normal text. It is used mostly in the web pages to generate the dynamic content such as writing the validation message, password strength etc.

Creating New HTML Elements (Nodes)

```
<div id="div1">  
  <p id="p1">This is a paragraph.</p>  
  <p id="p2">This is another paragraph.</p>  
</div>  
  
<script>  
var para = document.createElement("p");  
var node = document.createTextNode("This is new.");  
para.appendChild(node);  
var element = document.getElementById("div1");  
element.appendChild(para);  
</script>
```


Removing Existing HTML Elements

To remove an HTML element, use the `remove()` method:

Example

```
<div>  
  <p id="p1">This is a paragraph.</p>  
  <p id="p2">This is another paragraph.</p>  
</div>
```

```
<script>  
var elmnt = document.getElementById("p1");  
elmnt.remove();  
</script>
```

DOM Methods and Properties

DEMO-1

Validation

- Validations can be performed on the server side or on the client side (web browser).
- The user input validation take place on the Server Side during a post back session is called Server Side Validation.
- the user input validation take place on the Client Side (web browser) is called Client Side Validation. Client Side Validation does not require a post back.
- If the user request requires server resources to validate the user input, you should use Server Side Validation. If the user request does not require any server resources to validate the input, you can use Client Side Validation.

JavaScript Validation

DEMO-2